

第三节 两种基本积分法

3.1 换元积分法

1. 不定积分的换元法则(I)

定理3.1 设 $\int f(u)du = F(u) + C$ 具有原函数,
 $u = \varphi(x)$ 可导, 则有换元公式

$$\begin{aligned}\int f[\varphi(x)]\varphi'(x)dx &= \int f[\varphi(x)]d\varphi(x) \quad (\text{令 } \varphi(x) = u) \\ &= \int f(u)du \\ &= F(u) + C \\ &= F[\varphi(x)] + C.\end{aligned}$$

证 根据复合函数求导法则, 有

$$(F[\varphi(x)] + C)' = F'[\varphi(x)]\varphi'(x) = f[\varphi(x)]\varphi'(x),$$

这说明 $F[\varphi(x)] + C$ 是 $f[\varphi(x)]\varphi'(x)$ 的原函数.

例1 求 $\int \frac{x dx}{x^2 + 1}$.

解 因为

$$\int \frac{x dx}{x^2 + 1} = \frac{1}{2} \int \frac{2x dx}{x^2 + 1} = \frac{1}{2} \int \frac{d(x^2 + 1)}{x^2 + 1},$$

如果令 $u = x^2 + 1$, 则

$$\begin{aligned} \int \frac{x dx}{x^2 + 1} &= \frac{1}{2} \int \frac{du}{u} \\ &= \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln(1 + x^2) + C. \end{aligned}$$

例2 求 $\int x e^{x^2} dx$.

解

$$\begin{aligned}\int x e^{x^2} dx &= \frac{1}{2} \int e^{x^2} \cdot 2x dx \\ &= \frac{1}{2} \int e^{x^2} dx^2 = \frac{1}{2} e^{x^2} + C.\end{aligned}$$

在该例中, 实际上已经用了变量代换 $u = x^2$.

例3 求 $\int \frac{dx}{x(1+2\ln x)}$.

解

$$\begin{aligned}\int \frac{dx}{x(1+2\ln x)} &= \int \frac{d(\ln x)}{1+2\ln x} = \frac{1}{2} \int \frac{d(1+2\ln x)}{1+2\ln x} \\ &= \frac{1}{2} \ln|1+2\ln x| + C.\end{aligned}$$

在该例中, 实际上已经用了变量代换 $u = 1 + 2\ln x$.

例4 求 $\int \frac{dx}{e^x + 1}$.

解

$$\begin{aligned}\int \frac{dx}{e^x + 1} &= \int \frac{e^{-x} dx}{1 + e^{-x}} \\&= \int \frac{-de^{-x}}{1 + e^{-x}} \\&= - \int \frac{d(1 + e^{-x})}{1 + e^{-x}} \\&= -\ln|1 + e^{-x}| + C.\end{aligned}$$

在该例中, 实际上已经用了变量代换 $u = 1 + e^{-x}$.

例5 求 $\int \frac{dx}{x \cdot \ln x \cdot \ln(\ln x)}$.

解

$$\begin{aligned} \int \frac{dx}{x \cdot \ln x \cdot \ln(\ln x)} &= \int \frac{d(\ln x)}{\ln x \cdot \ln(\ln x)} \\ &= \int \frac{d\ln(\ln x)}{\ln(\ln x)} = \ln|\ln(\ln x)| + C. \end{aligned}$$

例6 求 $\int \frac{dx}{(\arcsin x)^2 \sqrt{1-x^2}}$.

解

$$\begin{aligned} \int \frac{dx}{(\arcsin x)^2 \sqrt{1-x^2}} &= \int \frac{d(\arcsin x)}{(\arcsin x)^2} \\ &= -\frac{1}{\arcsin x} + C. \end{aligned}$$

例7 设 $f(x)$ 在 $[0, 1]$ 上可微, $f(0) = 0$, 且 $\forall x \in (0, 1)$ 时,
 $f(x) \neq 0$, 试证: $\exists \xi \in (0, 1)$, 使得 $2\frac{f'(\xi)}{f(\xi)} = \frac{f'(1-\xi)}{f(1-\xi)}$.

解 定义

$$f(x) = 2\frac{f'(x)}{f(x)} - \frac{f'(1-x)}{f(1-x)}.$$

构造辅助函数, 使得 $F'(x) = f(x)$.

求不定积分,

$$\begin{aligned} F(x) &= \int f(x) dx = \int \left[2 \frac{f'(x)}{f(x)} - \frac{f'(1-x)}{f(1-x)} \right] dx \\ &= 2 \int \frac{f'(x)}{f(x)} dx - \int \frac{f'(1-x)}{f(1-x)} dx \\ &= 2 \int \frac{1}{f(x)} df(x) + \int \frac{f'(1-x)}{f(1-x)} d(1-x) \\ &= 2 \ln f(x) + \int \frac{1}{f(1-x)} df(1-x) \\ &= 2 \ln f(x) + \ln f(1-x), (+C \text{ 就不要了}). \end{aligned}$$

由题条件 $f(0) = 0$, 所以 $\lim_{x \rightarrow 0^+} F(x) = -\infty$,

$\lim_{x \rightarrow 1^-} F(x) = -\infty$. 于是 $\exists x_1, x_2 \in (0, 1)$, 使

得 $F(x_1) = F(x_2)$. 再由罗尔定理, 得结论成立.

⊙ 几种特殊类型的不定积分

例1 求 $\int \frac{dx}{a^2 + x^2}$.

解

$$\begin{aligned}\int \frac{dx}{a^2 + x^2} &= \int \frac{dx}{a^2 \left[1 + \left(\frac{x}{a} \right)^2 \right]} \\ &= \frac{1}{a} \int \frac{d\left(\frac{x}{a}\right)}{1 + \left(\frac{x}{a}\right)^2} = \frac{1}{a} \arctan \frac{x}{a} + C.\end{aligned}$$

例2 求 $\int \frac{dx}{a^2 - x^2} (a \neq 0)$.

解

$$\begin{aligned}\int \frac{dx}{a^2 - x^2} &= \int \frac{dx}{(a - x)(a + x)} \\&= \frac{1}{2a} \int \left[\frac{1}{a + x} + \frac{1}{a - x} \right] dx \\&= \frac{1}{2a} \int \frac{d(a + x)}{a + x} - \frac{1}{2a} \int \frac{d(a - x)}{a - x} \\&= \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right| + C.\end{aligned}$$

例3 求 $\int \frac{dx}{2+3x^2}, \quad \int \frac{dx}{2-3x^2}.$

解

$$\begin{aligned}\int \frac{dx}{2+3x^2} &= \frac{1}{\sqrt{3}} \int \frac{d\sqrt{3}x}{(\sqrt{2})^2 + (\sqrt{3}x)^2} \\ &= \frac{1}{\sqrt{6}} \arctan \frac{\sqrt{3}x}{\sqrt{2}} + C.\end{aligned}$$

$$\begin{aligned}\int \frac{dx}{2-3x^2} &= \frac{1}{\sqrt{3}} \int \frac{d\sqrt{3}x}{(\sqrt{2})^2 - (\sqrt{3}x)^2} \\ &= \frac{1}{2\sqrt{6}} \ln \left| \frac{\sqrt{2} + \sqrt{3}x}{\sqrt{2} - \sqrt{3}x} \right| + C.\end{aligned}$$

例4 求 $\int \frac{dx}{x^2 + 2x + 3}$.

解

$$\begin{aligned}\int \frac{dx}{x^2 + 2x + 3} &= \int \frac{dx}{(x + 1)^2 + 2} \\ &= \int \frac{d(x + 1)}{(x + 1)^2 + (\sqrt{2})^2} \\ &= \frac{1}{\sqrt{2}} \arctan \frac{x + 1}{\sqrt{2}} + C.\end{aligned}$$

例5 求 $\int \frac{dx}{-x^2 + 2x + 3}$.

解

$$\begin{aligned}\int \frac{dx}{-x^2 + 2x + 3} &= \int \frac{dx}{4 - (x - 1)^2} \\ &= \int \frac{d(x - 1)}{2^2 - (x - 1)^2} \\ &= \frac{1}{4} \ln \left| \frac{2 + (x - 1)}{2 - (x - 1)} \right| + C.\end{aligned}$$

例6 求 $\int \frac{dx}{Ax^2 + Bx + C}$.

解 把 $Ax^2 + Bx + C$ 配方, 化为 $\int \frac{dx}{a^2 + x^2}$ 或 $\int \frac{dx}{a^2 - x^2}$

例7 求 $\int \frac{x dx}{x^2 + 2x + 3}$.

解

$$\begin{aligned}\int \frac{x dx}{x^2 + 2x + 3} &= \frac{1}{2} \int \frac{(2x + 2 - 2) dx}{x^2 + 2x + 3} \\&= \frac{1}{2} \int \frac{(2x + 2) dx}{x^2 + 2x + 3} - \int \frac{dx}{x^2 + 2x + 3} \\&= \frac{1}{2} \int \frac{d(x^2 + 2x + 3)}{x^2 + 2x + 3} - \frac{1}{\sqrt{2}} \arctan \frac{x + 1}{\sqrt{2}} \\&= \frac{1}{2} \ln(x^2 + 2x + 3) - \frac{1}{\sqrt{2}} \arctan \frac{x + 1}{\sqrt{2}} + C.\end{aligned}$$

例8 求 $\int \frac{Px + Q}{Ax^2 + Bx + C} dx$.

解 因为 $(Ax^2 + Bx + C)' = 2Ax + B$, 所以

$$\begin{aligned} \int \frac{Px + Q}{Ax^2 + Bx + C} dx &= \int \frac{\frac{P}{2A}(2Ax + B) + Q - \frac{PB}{2A}}{Ax^2 + Bx + C} dx \\ &= \frac{P}{2A} \int \frac{(2Ax + B)dx}{Ax^2 + Bx + C} + \left(Q - \frac{PB}{2A}\right) \int \frac{dx}{Ax^2 + Bx + C} \\ &= \frac{P}{2A} \int \frac{d(Ax^2 + Bx + C)}{Ax^2 + Bx + C} + \left(Q - \frac{PB}{2A}\right) \int \frac{dx}{Ax^2 + Bx + C} \\ &= \frac{P}{2A} \ln|Ax^2 + Bx + C| + \left(Q - \frac{PB}{2A}\right) \int \frac{dx}{Ax^2 + Bx + C}. \end{aligned}$$

例9 求 $\int \frac{x dx}{4 + x^4}$.

解

$$\begin{aligned}\int \frac{x dx}{4 + x^4} &= \frac{1}{2} \int \frac{dx^2}{4 + x^4} \\ &= \frac{1}{2} \int \frac{dx^2}{2^2 + (x^2)^2} = \frac{1}{4} \arctan \frac{x^2}{2} + C.\end{aligned}$$

例10 求 $\int \frac{dx}{\sqrt{x}(1+x)}$.

解 因为 $d\sqrt{x} = \frac{dx}{2\sqrt{x}}$, 所以

$$\int \frac{dx}{\sqrt{x}(1+x)} = 2 \int \frac{d\sqrt{x}}{1+(\sqrt{x})^2} = 2\arctan\sqrt{x} + C.$$

例11 求 $\int \frac{dx}{\sqrt{a^2 - x^2}} (a > 0)$.

解

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \int \frac{d\left(\frac{x}{a}\right)}{\sqrt{1 - \left(\frac{x}{a}\right)^2}} = \arcsin \frac{x}{a} + C.$$

例12 求 $\int \frac{dx}{\sqrt{x(1-x)}}.$

解

$$\begin{aligned}\int \frac{dx}{\sqrt{x(1-x)}} &= \int \frac{dx}{\sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2}} \\&= \int \frac{d\left(x - \frac{1}{2}\right)}{\sqrt{\left(\frac{1}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2}} \\&= \arcsin(2x - 1) + C.\end{aligned}$$

又解

$$\begin{aligned}\int \frac{dx}{\sqrt{x(1-x)}} &= \int \frac{dx}{\sqrt{x} \cdot \sqrt{1-x}} \\ &= 2 \int \frac{d\sqrt{x}}{\sqrt{1-(\sqrt{x})^2}} = 2\arcsin\sqrt{x} + C.\end{aligned}$$

⊙ 三角函数的积分

例1 求 $\int \sin^3 x dx$.

解

$$\begin{aligned}\int \sin^3 x \, dx &= \int \sin^2 x \sin x \, dx \\&= - \int (1 - \cos^2 x) d(\cos x) \\&= - \int d(\cos x) + \int \cos^2 x d(\cos x) \\&= -\cos x + \frac{1}{3} \cos^3 x + C.\end{aligned}$$

例2 求 $\int \sin^2 x \cos^3 x \mathrm{d}x$.

解

$$\begin{aligned}\int \sin^2 x \cos^3 x \mathrm{d}x &= \int \sin^2 x \cos^2 x \cos x \mathrm{d}x \\&= \int \sin^2 x (1 - \sin^2 x) \mathrm{d}\sin x \\&= \int u^2 (1 - u^2) \mathrm{d}u = \cdots \\&\quad (\text{令 } u = \sin x).\end{aligned}$$

注1: 一般的, 对于

$$\sin^{2k+1} x \cos^n x \quad \text{或} \quad \sin^n x \cos^{2k+1} x$$

型函数的积分, 可做变换

$$u = \cos x, \quad \text{或} \quad u = \sin x.$$

例3 求 $\int \cos^2 x \mathrm{d}x$.

解

$$\begin{aligned}\int \cos^2 x \mathrm{d}x &= \int \frac{1 + \cos 2x}{2} \mathrm{d}x \\&= \frac{1}{2} \left(\int \mathrm{d}x + \int \cos 2x \mathrm{d}x \right) \\&= \frac{1}{2} \int \mathrm{d}x + \frac{1}{4} \int \cos 2x \mathrm{d}(2x) \\&= \frac{x}{2} + \frac{\sin 2x}{4} + C.\end{aligned}$$

例4 求 $\int \sin^2 x \cos^4 x \mathrm{d}x$.

解

$$\begin{aligned}\text{原式} &= \frac{1}{8} \int (1 - \cos 2x)(1 + \cos 2x)^2 \mathrm{d}x \\&= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) \mathrm{d}x \\&= \frac{1}{8} \int (\cos 2x - \cos^3 2x) \mathrm{d}x + \frac{1}{8} \int (1 - \cos^2 2x) \mathrm{d}x \\&= \frac{1}{8} \int \sin^2 2x \cdot \frac{1}{2} \mathrm{d}(\sin 2x) + \frac{1}{8} \int \frac{1}{2} (1 - \cos 4x) \mathrm{d}x \\&= \frac{1}{48} \sin^3 2x + \frac{x}{16} - \frac{1}{64} \sin 4x + C.\end{aligned}$$

注2: 一般的, 对于

$$\sin^{2k} x \cos^{2l} x$$

型函数的积分, 可利用三角恒等式

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x), \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x),$$

化为 $\cos 2x$ 的多项式. 然后用上例的方法求积分.

例5 求 $\int \sin 5x \cos 3x dx$.

解

$$\begin{aligned}\int \sin 5x \cos 3x dx &= \frac{1}{2} \int (\sin 8x + \sin 2x) dx \\&= \frac{1}{2} \left[\frac{1}{8} \int \sin 8x d(8x) + \frac{1}{2} \int \sin 2x d(2x) \right] \\&= -\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x + C.\end{aligned}$$

例6 求 $\int \frac{dx}{\cos x}$.

解 方法一

$$\begin{aligned}\int \frac{dx}{\cos x} &= \int \frac{\cos x dx}{\cos^2 x} \\ &= \int \frac{d\sin x}{1 - \sin^2 x} = \frac{1}{2} \ln \left| \frac{1 + \sin x}{1 - \sin x} \right| + C.\end{aligned}$$

方法二

$$\begin{aligned}\int \frac{dx}{\cos x} &= \int \frac{dx}{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}} \\&= 2 \int \frac{d(\frac{x}{2})}{\cos^2 \frac{x}{2} (1 - \tan^2 \frac{x}{2})} \\&= 2 \int \frac{d \tan(\frac{x}{2})}{1 - \tan^2 \frac{x}{2}} = \ln \left| \frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right| + C.\end{aligned}$$

2. 不定积分的换元规则(II)

定理3.2 设 f 是连续函数, $\varphi(t)$ 有连续导数, 且 $\varphi'(t)$ 定号, 则

$$\begin{aligned}\int f(x)dx &= \int f[\varphi(t)]\varphi'(t)dt \\ &= F(t) + C = F[\varphi^{-1}(x)] + C.\end{aligned}$$

其中, $\varphi^{-1}(x)$ 为 $\varphi(t)$ 的反函数.

证 因为 $(F(t))' = f[\varphi(t)]\varphi'(t)$, 则

$$\begin{aligned}(F[\varphi^{-1}(x)] + C)' &= f[\varphi(t)]\varphi'(t) \cdot (\varphi^{-1}(x))' \\ &= f[\varphi(t)]\varphi'(t) \cdot \frac{1}{\varphi'(t)} \\ &= f[\varphi(t)] = f(x).\end{aligned}$$

例1 求 $\int \sqrt{a^2 - x^2} dx$ ($a > 0$)

分析 求这个积分的困难在于有根式 $\sqrt{a^2 - x^2}$, 可利用三角公式

$$\sin^2 t + \cos^2 t = 1$$

化去根式 $\sqrt{a^2 - x^2}$.

解 设 $x = a \sin t$ ($-\frac{\pi}{2} < t < \frac{\pi}{2}$), 则 $dx = a \cos t dt$, 则

$$\begin{aligned} \int \sqrt{a^2 - x^2} dx &= \int a \cos t \cdot a \cos t dt = a^2 \int \cos^2 t dt \\ &= a^2 \int \frac{1 + \cos 2t}{2} dt \\ &= \frac{a^2}{2} t + \frac{a^2}{2} \sin t \cos t + C \end{aligned}$$

根据变换 $x = a \sin t$, 则有

$$\cos t = \sqrt{1 - \sin^2 t} = \sqrt{1 - \frac{x^2}{a^2}} = \frac{\sqrt{a^2 - x^2}}{a}.$$

于是

$$\int \sqrt{a^2 - x^2} dx = \frac{a^2}{2} \arcsin \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} + C.$$

例2 求 $\int \frac{dx}{\sqrt{x^2 + a^2}} \quad (a > 0)$

分析 求这个积分的困难在于有根式 $\sqrt{x^2 + a^2}$, 可利用三角公式

$$1 + \tan^2 t = \sec^2 t$$

来化去根式 $\sqrt{x^2 + a^2}$.

解 设 $x = a \tan t \left(-\frac{\pi}{2} < t < \frac{\pi}{2} \right)$, 则 $dx = a \sec^2 t dt$, 于是

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 + a^2}} &= \int \frac{a \sec^2 t}{a \sec t} dt \\ &= \int \sec t dt = \ln |\sec t + \tan t| + C. \end{aligned}$$

根据变换 $x = a \tan t$, 并注意到 $t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ 时,
 $\sec t > 0$, 故

$$\sec t = \sqrt{1 + \tan^2 t} = \frac{\sqrt{x^2 + a^2}}{a},$$

因此

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 + a^2}} &= \ln \left(\frac{x}{a} + \frac{\sqrt{x^2 + a^2}}{a} \right) + C \\ &= \ln(x + \sqrt{x^2 + a^2}) + C \end{aligned}$$

例3 求 $\int \frac{dx}{\sqrt{x^2 - a^2}} \quad (a > 0, |x| > a).$

分析 求这个积分的困难在于有根式 $\sqrt{x^2 - a^2}$, 可利用三角公式

$$\sec^2 t - 1 = \tan^2 t$$

来化去根式 $\sqrt{x^2 - a^2}$.

解 被积函数的定义域是 $(-\infty, -a) \cup (a, +\infty)$.

先求被积函数在 $(a, +\infty)$ 内的不定积分.

令 $x = a \sec t \quad (0 < t < \frac{\pi}{2})$, 则

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 - a^2}} &= \int \frac{a \sec t \cdot \tan t}{a \tan t} dt = \int \frac{dt}{\cos t} \\ &= \ln |\sec t + \tan t| + C \end{aligned}$$

再求被积函数在 $(-\infty, -a)$ 内的不定积分.

令 $u = -x$, 则 $u \in (a, +\infty)$, 且

$$\begin{aligned} \int \frac{dx}{\sqrt{x^2 - a^2}} &= - \int \frac{du}{\sqrt{u^2 - a^2}} \\ &= -\ln(u + \sqrt{u^2 - a^2}) + C = -\ln(-x + \sqrt{x^2 - a^2}) + C \\ &= -\ln \left[\frac{(-x + \sqrt{x^2 - a^2})(-x - \sqrt{x^2 - a^2})}{-x - \sqrt{x^2 - a^2}} \right] + C \\ &= -\ln \left[\frac{a^2}{-x - \sqrt{x^2 - a^2}} \right] + C \\ &= \ln(-x - \sqrt{x^2 - a^2}) + C_1, \text{ 其中 } C_1 = -\ln a^2 + C. \end{aligned}$$

两个区间内的不定积分可统一表达为

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln|x + \sqrt{x^2 - a^2}| + C.$$

⊙总结:

(1) 如果被积函数含有 $\sqrt{a^2 - x^2}$, 可作代换 $x = a \sin t$ 化去根式;

(2) 如果被积函数含有 $\sqrt{x^2 + a^2}$, 可作代换 $x = a \tan t$ 化去根式;

(3) 如果被积函数含有 $\sqrt{x^2 - a^2}$, 可作代换 $x = a \sec t$ 化去根式.

基本积分表

$$(20) \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan \frac{x}{a} + C;$$

$$(21) \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C;$$

$$(22) \int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + C;$$

$$(23) \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln |x + \sqrt{x^2 \pm a^2}| + C.$$

⊙思考题:

求 $\int \frac{dx}{\sqrt{ax^2 + bx + c}}, \quad \int \frac{px + q}{\sqrt{ax^2 + bx + c}} dx?$

例4 求 $\int \frac{dx}{\sqrt{1+x-x^2}} dx$.

解

$$\begin{aligned} \int \frac{dx}{\sqrt{1+x-x^2}} dx &= \int \frac{d\left(x - \frac{1}{2}\right)}{\sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2}} dx \\ &= \arcsin \frac{2x-1}{\sqrt{5}}. \end{aligned}$$

例5 求 $\int \frac{1-x}{\sqrt{9-4x^2}} dx$.

解

$$\begin{aligned}\text{原式} &= \int \frac{1}{\sqrt{9-4x^2}} dx + \frac{1}{8} \int \frac{-8x}{\sqrt{9-4x^2}} dx \\ &= \frac{1}{2} \int \frac{1}{\sqrt{9-(2x)^2}} d(2x) + \frac{1}{8} \int \frac{1}{\sqrt{9-4x^2}} d(9-4x^2) \\ &= \frac{1}{2} \arcsin \frac{2x}{3} + \frac{1}{4} \sqrt{9-4x^2} + C.\end{aligned}$$

例6 求 $\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}.$

解 设 $\sqrt[6]{x} = t$, 则 $x = t^6, dx = 6t^5 dt$, 于是

$$\begin{aligned}\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}} &= \int \frac{6t^5 dt}{t^3 + t^2} \\&= 6 \int \frac{t^3 dt}{1+t} = 6 \int \frac{t^3 + 1 - 1}{1+t} dt \\&= 6 \int \left(t^2 - t + 1 - \frac{1}{1+t} \right) dt \\&= 2t^3 - 3t^2 + 6t - 6\ln|1+t| + C \\&= 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\ln|1 + \sqrt[6]{x}| + C.\end{aligned}$$

例7 求 $\int x^2 \sqrt[3]{1-x} dx$.

解 设 $1-x=t$, 则 $x=1-t$, $dx=-dt$, 于是

$$\begin{aligned} \text{原式} &= - \int (1-t)^2 t^{\frac{1}{3}} dt \\ &= - \int (t^{\frac{1}{3}} - 2t^{\frac{4}{3}} + t^{\frac{7}{3}}) dt \\ &= -\frac{3}{4}t^{\frac{4}{3}} + \frac{6}{7}t^{\frac{7}{3}} - \frac{3}{10}t^{\frac{10}{3}} + C \\ &= -\frac{3}{4}(1-x)^{\frac{4}{3}} + \frac{6}{7}(1-x)^{\frac{7}{3}} - \frac{3}{10}(1-x)^{\frac{10}{3}} + C. \end{aligned}$$

例8 求 $\int \cos^3 x \cdot \sqrt{\sin x} dx$.

解 设 $\sin x = t$, 则

$$\cos^3 x dx = (1 - \sin^2 x) d\sin x = (1 - t^2) dt,$$

于是

$$\begin{aligned} \int \cos^3 x \cdot \sqrt{\sin x} dx &= \int (1 - t^2) t^{\frac{1}{2}} dt \\ &= \frac{2}{3} t^{\frac{3}{2}} - \frac{2}{7} t^{\frac{7}{2}} + C \\ &= \frac{2}{3} (\sin x)^{\frac{3}{2}} - \frac{2}{7} (\sin x)^{\frac{7}{2}} + C. \end{aligned}$$

有理函数的不定积分

问题: $\int \frac{P(x)}{Q(x)} dx$, 其中

$$\frac{P(x)}{Q(x)} = \frac{a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n}{b_0x^m + b_1x^{m-1} + \cdots + b_{m-1}x + b_m}.$$

例1 $\int \frac{x^4 + 2x^2 - 1}{x^3 + 1} dx.$

方法: (共分三步)

步1: 将有理函数化成多项式与真分式之和:

$$\frac{P(x)}{Q(x)} = R(x) + \frac{P_1(x)}{Q(x)},$$

则

$$\int \frac{P(x)}{Q(x)} dx = \int R(x) dx + \int \frac{P_1(x)}{Q(x)} dx.$$

例1

$$\int \frac{x^4 + 2x^2 - 1}{x^3 + 1} dx.$$

解

$$\begin{aligned}\frac{x^4 + 2x^2 - 1}{x^3 + 1} &= \frac{x(x^3 + 1) + (2x^2 - x - 1)}{x^3 + 1} \\ &= x + \frac{2x^2 - x - 1}{x^3 + 1},\end{aligned}$$

于是

$$\begin{aligned}\int \frac{x^4 + 2x^2 - 1}{x^3 + 1} dx &= \int x dx + \int \frac{2x^2 - x - 1}{x^3 + 1} dx \\ &= \frac{x^2}{2} + \int \frac{2x^2 - x - 1}{x^3 + 1} dx.\end{aligned}$$

步2: 将真分式化为部分分式之和.

(1). 将 $Q(x)$ 在实数范围内分解成一次因式和二次因式的乘积.

$$Q(x) = b_0(x-a)^\alpha \cdots (x-b)^\beta (x^2+px+q)^\lambda \cdots (x^2+rx+s)^\mu$$

其中 $p^2 - 4q < 0, \cdots, r^2 - 4s < 0$.

(2). 将真分式拆成若干个部分分式之和

$$\begin{aligned}\frac{P_1(x)}{Q(x)} &= \frac{A_1}{(x-a)^\alpha} + \frac{A_2}{(x-a)^{\alpha-1}} + \cdots + \frac{A_\alpha}{(x-a)} + \cdots \\ &+ \frac{B_1}{(x-b)^\beta} + \frac{B_2}{(x-b)^{\beta-1}} + \cdots + \frac{B_\beta}{(x-b)} + \cdots \\ &+ \frac{M_1x + N_1}{(x^2 + px + q)^\lambda} + \frac{M_2x + N_2}{(x^2 + px + q)^{\lambda-1}} + \cdots \\ &+ \frac{M_\lambda x + N_\lambda}{(x^2 + px + q)} + \cdots + \frac{R_1x + S_1}{(x^2 + rx + s)^\mu} \\ &+ \frac{R_2x + S_2}{(x^2 + rx + s)^{\mu-1}} + \cdots + \frac{R_\mu x + S_\mu}{(x^2 + rx + s)}\end{aligned}$$

例1 (续) 因为 $Q(x) = (x+1)(x^2 - x + 1)$, 则

$$\begin{aligned}\frac{P_1(x)}{Q(x)} &= \frac{2x^2 - x - 1}{x^3 + 1} = \frac{2x^2 - x - 1}{(x+1)(x^2 - x + 1)} \\ &= \frac{A}{x+1} + \frac{Bx + C}{x^2 - x + 1},\end{aligned}$$

两端去分母, 得,

$$2x^2 - x - 1 = A(x^2 - x + 1) + (Bx + C)(x + 1)$$

取: $x = -1$, 则 $2 = A \cdot 3$, 得: $A = \frac{2}{3}$.

$x = 0$, 则 $-1 = A + C$, 得: $C = -\frac{5}{3}$.

$x = 1$, 则 $0 = A + 2(B + C)$, 得: $B = \frac{4}{3}$.

则

$$\frac{2x^2 - x - 1}{x^3 + 1} = \frac{2}{3(x+1)} + \frac{4x-5}{3(x^2-x+1)}.$$

于是

$$\begin{aligned} & \int \frac{x^4 + 2x^2 - 1}{x^3 + 1} dx \\ &= \int x dx + \int \frac{2x^2 - x - 1}{x^3 + 1} dx \\ &= \frac{x^2}{2} + \int \frac{2}{3(x+1)} dx + \int \frac{4x-5}{3(x^2-x+1)} dx. \end{aligned}$$

步3: 对部分分式进行积分.

称下列四种分式为最简真分式:

$$\frac{A}{x-a}, \quad \frac{A}{(x-a)^m} \quad (m > 1),$$

$$\frac{Bx+C}{x^2+px+q}, \quad \frac{Bx+C}{(x^2+px+q)^k} \quad (k > 1, p^2-4q < 0).$$

求有理函数的积分, 可以归结为求四个最简真分式的积分.

$$\int \frac{A}{x-a} dx = A \ln|x-a| + C;$$

$$\int \frac{A}{(x-a)^m} dx = \frac{A}{1-m} \cdot \frac{1}{(x-a)^{m-1}} + C;$$

以及

$$\begin{aligned}\int \frac{Bx + C}{(x^2 + px + q)^k} dx &= \int \frac{Bx + C}{\left[\left(x + \frac{p}{2}\right)^2 + q - \frac{p^2}{4}\right]^k} dx \\&= \int \frac{Bx + C}{\left[\left(x + \frac{p}{2}\right)^2 + a^2\right]^k} dx \quad \left(\text{令 } a = \sqrt{q - \frac{p^2}{4}}\right) \\&= \int \frac{Bt + C - \frac{Bp}{2}}{(t^2 + a^2)^k} dt \\&= B \int \frac{t}{(t^2 + a^2)^k} dt + \left(C - \frac{Bp}{2}\right) \int \frac{dt}{(t^2 + a^2)^k}.\end{aligned}$$

上式中的两个积分总可以通过下面四个积分求出:

$$\int \frac{t}{t^2 + a^2} dt = \frac{1}{2} \ln(t^2 + a^2) + C;$$

$$\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \arctan \frac{t}{a} + C;$$

$$\int \frac{t dt}{(t^2 + a^2)^k} = \frac{1}{2(1-k)} \cdot \frac{1}{(t^2 + a^2)^{k-1}} + C \quad (k > 1);$$

$$\int \frac{dt}{(t^2 + a^2)^k} \text{ 可利用210页例3.17求得.}$$

例1 (续)

$$\begin{aligned}& \int \frac{x^4 + 2x^2 - 1}{x^3 + 1} dx \\&= \int x dx + \int \frac{2}{3(x+1)} dx + \int \frac{4x-5}{3(x^2-x+1)} dx \\&= \frac{x^2}{2} + \frac{2}{3} \ln|x+1| + \int \frac{2(2x-1)-3}{3(x^2-x+1)} dx \\&= \frac{x^2}{2} + \frac{2}{3} \ln|x+1| + \frac{2}{3} \int \frac{d(x^2-x+1)}{x^2-x+1} dx - \int \frac{dx}{x^2-x+1} \\&= \frac{x^2}{2} + \frac{2}{3} \ln|x+1| + \frac{2}{3} \ln|x^2-x+1| - \int \frac{dx}{\left(x-\frac{1}{2}\right)^2 + \frac{3}{4}} \\&= \frac{x^2}{2} + \frac{2}{3} \ln|x+1| + \frac{2}{3} \ln|x^2-x+1| - \frac{2}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} + C\end{aligned}$$

例2 计算 $\int \frac{x^4 + 4x^3 + 11x^2 + 12x + 8}{x^3 + 3x^2 + 5x + 3} dx$.

解 因为

$$\begin{aligned}\text{被积函数} &= (x + 1) + \frac{3x^2 + 4x + 5}{x^3 + 3x^2 + 5x + 3} \\ &= (x + 1) + \frac{3x^2 + 4x + 5}{(x + 1)(x^2 + 2x + 3)} \\ &= (x + 1) + \frac{2}{x + 1} + \frac{x - 1}{x^2 + 2x + 3},\end{aligned}$$

$$\begin{aligned}
 & \text{则} \int \frac{x^4 + 4x^3 + 11x^2 + 12x + 8}{x^3 + 3x^2 + 5x + 3} dx \\
 &= \int (x + 1) dx + \int \frac{2}{x + 1} dx + \int \frac{x - 1}{x^2 + 2x + 3} dx \\
 &= \frac{(x + 1)^2}{2} + 2\ln|x + 1| + \frac{1}{2} \int \left(\frac{2x + 2}{x^2 + 2x + 3} - \frac{4}{x^2 + 2x + 3} \right) dx \\
 &= \frac{(x + 1)^2}{2} + 2\ln|x + 1| \\
 &\quad + \frac{1}{2} \int \frac{d(x^2 + 2x + 3)}{x^2 + 2x + 3} - 2 \int \frac{1}{(x + 1)^2 + 2} dx \\
 &= \frac{(x + 1)^2}{2} + 2\ln|x + 1| \\
 &\quad + \frac{1}{2} \ln(x^2 + 2x + 3) - \sqrt{2} \arctan \frac{x + 1}{\sqrt{2}} + C.
 \end{aligned}$$

三角有理函数的不定积分

问题: $\int R(\sin x, \cos x) dx$. 其中, $R(u, v)$ 表示 u, v 两个变量的有理式.

例如:

$$\sin^3 x, \frac{1}{\cos x}, \frac{1 + \sin x}{\cot x (1 + \sqrt{3} \cos x)}$$

等等, 都是三角有理函数; 而

$$\frac{\sin x \cos x}{\sqrt{\cos^2 x + 2 \sin^2 x}}$$

不是三角有理函数.

方法: 令 $\tan \frac{x}{2} = t$, 或 $x = 2\arctan t$, 在这个变换下, 三角有理函数的积分, 可化为关于 t 的有理函数的积分. 因

$$\sin x = \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{2t}{1 + t^2},$$

$$\cos x = \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{1 - t^2}{1 + t^2},$$

$$dx = \frac{2dt}{1 + t^2},$$

所以

$$\int R(\sin x, \cos x) dx = \int R\left(\frac{2t}{1 + t^2}, \frac{1 - t^2}{1 + t^2}\right) \frac{2dt}{1 + t^2}.$$

例1 求 $\int \frac{1}{3 + \cos x} dx$.

解 令 $t = \tan \frac{x}{2}$, 有 $dx = \frac{2}{1 + t^2} dt$, 于是

$$\begin{aligned}\int \frac{1}{3 + \cos x} dx &= \int \frac{1}{3 + \frac{1 - t^2}{1 + t^2}} \cdot \frac{2}{1 + t^2} dt \\&= \int \frac{1}{2 + t^2} dt = \frac{1}{\sqrt{2}} \arctan \frac{t}{\sqrt{2}} + C \\&= \frac{1}{\sqrt{2}} \arctan \left(\frac{1}{\sqrt{2}} \tan \frac{x}{2} \right) + C.\end{aligned}$$

例2 求 $\int \frac{1}{\sin x + \cos x} dx$.

解 令 $t = \tan \frac{x}{2}$, 有 $dx = \frac{2}{1+t^2} dt$, 于是

$$\begin{aligned} \text{原式} &= \int \frac{1}{\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\ &= \int \frac{2}{1+2t-t^2} dt = \frac{\sqrt{2}}{2} \ln \left| \frac{t - (1 - \sqrt{2})}{t - (1 + \sqrt{2})} \right| + C \\ &= \frac{\sqrt{2}}{2} \ln \left| \frac{\tan \frac{x}{2} - (1 - \sqrt{2})}{\tan \frac{x}{2} - (1 + \sqrt{2})} \right| + C. \end{aligned}$$

另外三种经常采用的变换形式.

情形1 当

$$R(-\sin x, \cos x) = -R(\sin x, \cos x)$$

时, 令 $t = \cos x$, 就可能转化为 t 的有理函数.

例如 $\int \frac{\sin^5 x}{\cos^4 x} dx$, 令 $t = \cos x$. 于是

$$\begin{aligned} \int \frac{\sin^5 x}{\cos^4 x} dx &= - \int \frac{\sin^4 x}{\cos^4 x} d\cos x = - \int \frac{(1 - \cos^2 x)^2}{\cos^4 x} d\cos x \\ &= - \int \frac{(1 - t^2)^2}{t^4} dt = - \int \left(\frac{1}{t^4} - \frac{2}{t^2} + 1 \right) dt. \end{aligned}$$

情形2 当

$$R(\sin x, -\cos x) = -R(\sin x, \cos x)$$

时, 令 $t = \sin x$, 就可能转化为 t 的有理函数.

例如

$$\begin{aligned}\int \frac{1}{\sin x \sin 2x} dx &= \frac{1}{2} \int \frac{1}{\sin^2 x \cos x} dx \\ &= \frac{1}{2} \int \frac{\cos x}{\sin^2 x \cos^2 x} dx \\ &= \frac{1}{2} \int \frac{d\sin x}{\sin^2 x (1 - \sin^2 x)} = \frac{1}{2} \int \frac{1}{t^2 (1 - t^2)} dt.\end{aligned}$$

情形3 当

$$R(-\sin x, -\cos x) = R(\sin x, \cos x)$$

时, 令 $t = \tan x$, 就可能转化为 t 的有理函数.

例如

$$\begin{aligned}\int \frac{dx}{a^2 \sin^2 x + b^2 \cos^2 x} &= \int \frac{dx}{\cos^2 x \cdot (a^2 \tan^2 x + b^2)} \\ &= \int \frac{d \tan x}{a^2 \tan^2 x + b^2} = \int \frac{dt}{a^2 t^2 + b^2}.\end{aligned}$$

例3 求 $\int \tan^3 x dx$.

方法一: 不定积分属于情形1, 所以

$$\begin{aligned}\int \tan^3 x dx &= \int \frac{\sin^3 x}{\cos^3 x} dx = - \int \frac{\sin^2 x}{\cos^3 x} d\cos x \\&= - \int \frac{1 - \cos^2 x}{\cos^3 x} d\cos x \\&= - \int \frac{1 - t^2}{t^3} dt \quad (\text{令 } t = \cos x) \\&= \frac{1}{2} \frac{1}{t^2} + \ln|t| + C \\&= \frac{1}{2} \frac{1}{\cos^2 x} + \ln|\cos x| + C.\end{aligned}$$

方法二：不定积分也属于情形2，所以

$$\begin{aligned}\int \tan^3 x dx &= \int \frac{\sin^3 x \cos x}{\cos^4 x} dx = \int \frac{\sin^3 x}{\cos^4 x} d\sin x \\&= \int \frac{\sin^3 x}{(1 - \sin^2 x)^2} d\sin x \\&= \int \frac{t^3}{(1 - t^2)^2} dt \quad (\text{令 } t = \sin x) \\&\dots\dots\dots\end{aligned}$$

方法三：不定积分也属于情形3，所以

$$\begin{aligned}\int \tan^3 x dx &= \int \frac{\sin^3 x}{\cos x \cos^2 x} dx = \int \tan x \sin^2 x d \tan x \\&= \int \tan x (1 - \cos^2 x) d \tan x \\&= \int \tan x \left(1 - \frac{1}{1 + \tan^2 x} \right) d \tan x \\&= \int t \left(1 - \frac{1}{1 + t^2} \right) dt \quad (\text{令 } t = \tan x) \\&= \frac{1}{2} t^2 - \frac{1}{2} \ln(1 + t^2) + C \\&= \frac{1}{2} \tan^2 x - \frac{1}{2} \ln(1 + \tan^2 x) + C.\end{aligned}$$

例4 求 $\int \frac{dx}{2\cos^2x + \sin x \cos x + \sin^2x}.$

解

$$\begin{aligned} & \int \frac{dx}{2\cos^2x + \sin x \cos x + \sin^2x} \\ &= \int \frac{dx}{\cos^2x(2 + \tan x + \tan^2x)} \\ &= \int \frac{d\tan x}{2 + \tan x + \tan^2x} \\ &= \int \frac{dt}{2 + t + t^2} \quad (\text{令 } t = \tan x) \\ & \dots\dots\dots \end{aligned}$$

例5 求 $\int \frac{\sin^2 x}{1 + \cos^2 x} dx$.

解

$$\begin{aligned}\int \frac{\sin^2 x}{1 + \cos^2 x} dx &= \int \frac{1 - \cos^2 x}{\cos^2 x (\sec^2 x + 1)} dx \\&= \int \frac{1 - \cos^2 x}{\sec^2 x + 1} d(\tan x) \\&= \int \frac{1 - \frac{1}{1 + \tan^2 x}}{1 + \tan^2 x + 1} d(\tan x) \\&= \int \frac{t^2}{(2 + t^2)(1 + t^2)} dt \quad (\text{令 } t = \tan x) \\&\dots\dots\dots\end{aligned}$$

某些含有根式的积分

类型一

$$\int R \left(x, \sqrt[n]{\frac{ax+b}{cx+d}} \right) dx,$$

方法:

可令 $\frac{ax+b}{cx+d} = t^n$, 即可化为有理函数的积分;

例1 求 $\int \frac{1}{x} \sqrt{\frac{x+1}{x}} dx$.

解 令 $t = \sqrt{\frac{x+1}{x}}$, $t^2 = \frac{x+1}{x}$, 有 $x = \frac{1}{t^2-1}$, 则

$dx = -\frac{2t}{(t^2-1)^2} dt$, 于是

$$\begin{aligned} \int \frac{1}{x} \sqrt{\frac{x+1}{x}} dx &= \int (t^2-1) \cdot t \cdot \left[-\frac{2t}{(t^2-1)^2} \right] dt \\ &= -2 \int \frac{t^2}{t^2-1} dt = -2t - \ln \left| \frac{t-1}{t+1} \right| + C \\ &= -2\sqrt{\frac{x+1}{x}} - \ln \left| \left(\sqrt{\frac{x+1}{x}} - 1 \right) / \left(\sqrt{\frac{x+1}{x}} + 1 \right) \right| \\ &\quad + C. \end{aligned}$$

某些含有根式的积分

类型二

$$\int R(x, \sqrt{ax^2 + bx + c}) dx.$$

方法:

可先将根式内的二次三项式配方, 然后作三角代换即可将该积分化为三角有理函数的积分.

例2 求 $\int \frac{x^2}{(4-2x+x^2)\sqrt{2+2x-x^2}} dx$.

解 因为

$$\frac{x^2}{(4-2x+x^2)\sqrt{2+2x-x^2}} = \frac{(x-1)^2 + 2(x-1) + 1}{[3+(x-1)^2]\sqrt{3-(x-1)}}$$

令 $x-1 = \sqrt{3}\sin t$ $\left(-\frac{\pi}{2} < t < \frac{\pi}{2}\right)$, 则

$$\sqrt{2+2x-x^2} = \sqrt{3}\cos t, dx = \sqrt{3}\cos t dt,$$

于是

$$\begin{aligned}& \int \frac{x^2}{(4-2x+x^2)\sqrt{2+2x-x^2}} dx \\&= \int \frac{1+2\sqrt{3}\sin t+3\sin^2 t}{3(1+\sin^2 t)} dt \\&= \int dt - \frac{2}{\sqrt{3}} \int \frac{d\cos t}{2-\cos^2 t} - \frac{2}{3} \int \frac{dt \tan t}{1+2\tan^2 t} \\&= t - \frac{1}{\sqrt{6}} \ln \left| \frac{\sqrt{2} + \cos t}{\sqrt{2} - \cos t} \right| - \frac{\sqrt{2}}{3} \arctan(\sqrt{2} \tan t) + C \\&= \arcsin \frac{x-1}{\sqrt{3}} - \frac{1}{\sqrt{6}} \ln \left| \frac{\sqrt{6} + \sqrt{2+2x-x^2}}{\sqrt{6} - \sqrt{2+2x-x^2}} \right| \\&\quad - \frac{\sqrt{2}}{3} \arctan \frac{(x-1)\sqrt{2}}{\sqrt{2+2x-x^2}} + C.\end{aligned}$$

3. 定积分的换元法

定理3.3 设 $f(x) \in C[a, b]$. 如果函数 $x = \varphi(t)$ 满足

(1) $\varphi(\alpha) = a, \varphi(\beta) = b$;

(2) $\varphi'(t) \in C[\alpha, \beta]$ (或 $\varphi'(t) \in C[\beta, \alpha]$), 且其值域 $R_\varphi = [a, b]$

则有

$$\int_a^b f(x) dx = \int_\alpha^\beta f[\varphi(t)] \varphi'(t) dt.$$

证 上式两端的被积函数都是连续函数, 故上式两端的定积分一定存在.

设 $F(x)$ 是 $f(x)$ 的一个原函数, 则

$$\int_a^b f(x)dx = F(b) - F(a).$$

另一方面, 对于复合函数 $F[\varphi(t)]$, 因为

$$\frac{dF[\varphi(t)]}{dt} = \frac{dF}{dx} \cdot \frac{dx}{dt} = f(x)\varphi'(t) = f[\varphi(t)]\varphi'(t),$$

即 $F[\varphi(t)]$ 是 $f[\varphi(t)]\varphi'(t)$ 的一个原函数. 因此

$$\int_{\alpha}^{\beta} f[\varphi(t)]\varphi'(t)dt = F[\varphi(\beta)] - F[\varphi(\alpha)] = F(b) - F(a).$$

得证.

例1 求 $\int_0^1 x^2 \sqrt{1-x^2} dx$.

解 设 $x = \sin t, 0 \leq t \leq \frac{\pi}{2}$, 则 $dx = \cos t dt$.

且当 $x = 0$ 时, $t = 0$; 当 $x = 1$ 时, $t = \frac{\pi}{2}$.

利用定理可得

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \sin^2 t \cos^2 t dt = \frac{1}{4} \int_0^{\frac{\pi}{2}} \sin^2 2t dt \\ &= \frac{1}{8} \int_0^{\frac{\pi}{2}} (1 - \cos 4t) dt \\ &= \frac{1}{8} \left[t - \frac{\sin 4t}{4} \right] \bigg|_0^{\frac{\pi}{2}} = \frac{\pi}{16}. \end{aligned}$$

例2 求 $\int_a^{2a} \frac{\sqrt{x^2 - a^2}}{x^4} dx, \quad (a > 0)$

解 设 $x = a \csc t$, 即 $\sin t = \frac{a}{x}$.

则当 $x = 2a$ 时, $t = \frac{\pi}{6}$; 当 $x = a$ 时, $t = \frac{\pi}{2}$.

于是

$$\begin{aligned} \int_a^{2a} \frac{\sqrt{x^2 - a^2}}{x^4} dx &= \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\sqrt{a^2 \csc^2 t - a^2}}{a^4 \csc^4 t} da \csc t \\ &= \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \left(-\frac{1}{a^2} \cos^2 t \cdot \sin t \right) dt \\ &= \frac{1}{3a^2} \cos^3 t \Big|_{\frac{\pi}{2}}^{\frac{\pi}{6}} = \frac{\sqrt{3}}{8a^2}. \end{aligned}$$

换元公式也可以反过来使用, 即

$$\int_{\alpha}^{\beta} f[\varphi(t)]\varphi'(t)\mathrm{d}t = \int_a^b f(x)\mathrm{d}x.$$

例3 计算积分 $\int_0^{\pi} \sqrt{\sin x - \sin^3 x} dx$.

解

$$\begin{aligned} \text{原式} &= \int_0^{\pi} \sqrt{\sin x} \sqrt{1 - \sin^2 x} dx \\ &= \int_0^{\pi} \sqrt{\sin x} |\cos x| dx \\ &= \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \cos x dx - \int_{\frac{\pi}{2}}^{\pi} \sqrt{\sin x} \cos x dx \\ &= \int_0^{\frac{\pi}{2}} \sqrt{\sin x} d\sin x - \int_{\frac{\pi}{2}}^{\pi} \sqrt{\sin x} d\sin x \\ &= \frac{2}{3} \sin^{\frac{3}{2}} x \Big|_0^{\frac{\pi}{2}} - \frac{2}{3} \sin^{\frac{3}{2}} x \Big|_{\frac{\pi}{2}}^{\pi} = \frac{4}{3}. \end{aligned}$$

例4 求 $\int_0^{\pi} \sqrt{\sin^3 x - \sin^5 x} dx$.

解 由于 $\sqrt{\sin^3 x - \sin^5 x} = \sin^{\frac{3}{2}} x \cdot |\cos x|$.

当 $x \in \left[0, \frac{\pi}{2}\right]$ 时, $|\cos x| = \cos x$; 当 $x \in \left[\frac{\pi}{2}, \pi\right]$ 时, $|\cos x| = -\cos x$. 所以

$$\begin{aligned} \text{原式} &= \int_0^{\frac{\pi}{2}} \sin^{\frac{3}{2}} x \cos x dx + \int_{\frac{\pi}{2}}^{\pi} \sin^{\frac{3}{2}} x (-\cos x) dx \\ &= \int_0^{\frac{\pi}{2}} \sin^{\frac{3}{2}} x d(\sin x) - \int_{\frac{\pi}{2}}^{\pi} \sin^{\frac{3}{2}} x d(\sin x) \\ &= \int_0^1 u^{\frac{3}{2}} du - \int_1^0 u^{\frac{3}{2}} du \quad (\text{令 } u = \sin x) \\ &= \left[\frac{2}{5} \sin^{\frac{5}{2}} x \right] \bigg|_0^{\frac{\pi}{2}} - \left[\frac{2}{5} \sin^{\frac{5}{2}} x \right] \bigg|_{\frac{\pi}{2}}^{\pi} = \frac{4}{5}. \end{aligned}$$

注: 在使用换元法计算定积分时, 要注意定理中对 $\varphi(t)$ 的要求.

例如, 计算定积分 $\int_0^{\pi} \frac{1}{1 + \sin^2 x} dx$.

因为

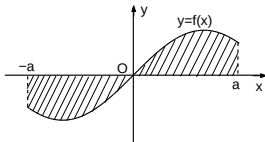
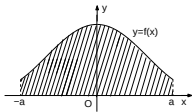
$$\begin{aligned}\int \frac{1}{1 + \sin^2 x} dx &= \int \frac{1}{1 + 2 \tan^2 x} d(\tan x) \quad (\text{令 } t = \tan x) \\ &= \frac{1}{\sqrt{2}} \arctan(\sqrt{2} \tan x) + C,\end{aligned}$$

如果对于定积分, 也使用同样的换元法, 令 $t = \tan x$ 时, 下述计算过程是错误的

$$\begin{aligned}\int_0^{\pi} \frac{1}{1 + \sin^2 x} dx &= \int_0^{\pi} \frac{1}{1 + 2 \tan^2 x} d(\tan x) \\ &= \int_0^0 \frac{1}{1 + 2t^2} dt = 0.\end{aligned}$$

例5 (1) 若 $f(x)$ 是 $[-a, a]$ 上可积的偶函数, 则

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx;$$



(2) 若 $f(x)$ 是 $[-a, a]$ 上可积的奇函数, 则

$$\int_{-a}^a f(x) dx = 0;$$

证 因为

$$\int_{-a}^a f(x) \mathrm{d}x = \int_{-a}^0 f(x) \mathrm{d}x + \int_0^a f(x) \mathrm{d}x,$$

对积分 $\int_{-a}^0 f(x) \mathrm{d}x$ 作变换 $x = -t$, 有 $\mathrm{d}x = -\mathrm{d}t$. 于是

$$\int_{-a}^0 f(x) \mathrm{d}x = - \int_a^0 f(-t) \mathrm{d}t = \int_0^a f(-x) \mathrm{d}x$$

则

$$\int_{-a}^a f(x) \mathrm{d}x = \int_0^a [f(x) + f(-x)] \mathrm{d}x.$$

(1) 若 $f(x)$ 为偶函数, 则

$$f(x) + f(-x) = 2f(x),$$

从而

$$\int_{-a}^a f(x) \mathrm{d}x = 2 \int_0^a f(x) \mathrm{d}x;$$

(2) 若 $f(x)$ 为奇函数, 则

$$f(x) + f(-x) = 0,$$

从而

$$\int_{-a}^a f(x) \mathrm{d}x = 0;$$

例5(b) 求 $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\sin x}{x^2 + 1} dx$.

解 因 $\frac{\sin x}{x^2 + 1}$ 是奇函数, 则

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{\sin x}{x^2 + 1} dx = 0.$$

例5(c). 求积分 $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + e^{-x}} dx$.

解 根据

$$\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx,$$

所以

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + e^{-x}} dx &= \int_0^{\frac{\pi}{2}} \left(\frac{\sin^2 x}{1 + e^{-x}} + \frac{\sin^2 x}{1 + e^x} \right) dx \\ &= \int_0^{\frac{\pi}{2}} \sin^2 x dx = \frac{\pi}{4}. \end{aligned}$$

例6 若 $f(x) \in C[0, 1]$, 证明

$$(1) \int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx;$$

$$(2) \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$$

由此计算

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx.$$

证 (1) 设 $x = \frac{\pi}{2} - t$, 有 $dx = -dt$.

当 $x = 0$ 时, $t = \frac{\pi}{2}$; 当 $x = \frac{\pi}{2}$ 时, $t = 0$.

于是

$$\begin{aligned}\int_0^{\frac{\pi}{2}} f(\sin x) dx &= - \int_{\frac{\pi}{2}}^0 f \left[\sin \left(\frac{\pi}{2} - t \right) \right] dt \\ &= \int_0^{\frac{\pi}{2}} f(\cos t) dt = \int_0^{\frac{\pi}{2}} f(\cos x) dx.\end{aligned}$$

(2) 设 $x = \pi - t$, 有 $dx = -dt$.

当 $x = 0$ 时, $t = \pi$; 当 $x = \pi$ 时, $t = 0$.

于是

$$\begin{aligned}\int_0^{\pi} x f(\sin x) dx &= - \int_{\pi}^0 (\pi - t) f[\sin(\pi - t)] dt \\&= \int_0^{\pi} (\pi - t) f(\sin t) dt \\&= \pi \int_0^{\pi} f(\sin t) dt - \int_0^{\pi} t f(\sin t) dt \\&= \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx\end{aligned}$$

移项整理得
$$\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$$

利用上述, 得

$$\begin{aligned}\int_0^{\pi} \frac{x \sin x}{2 - \sin^2 x} dx &= \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{2 - \sin^2 x} dx \\&= \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \\&= -\frac{\pi}{2} \int_0^{\pi} \frac{d \cos x}{1 + \cos^2 x} \\&= -\frac{\pi}{2} \arctan(\cos x) \Big|_0^{\pi} = \frac{\pi^2}{4}\end{aligned}$$

例7. 设 $f(x), g(x)$ 在 $[-a, a] (a > 0)$ 上连续, $g(x)$ 为偶函数, 且 $f(x)$ 满足条件

$$f(x) + f(-x) = A$$

其中, A 为常数.

(1). 证明:
$$\int_{-a}^a f(x)g(x)dx = A \int_0^a g(x)dx;$$

(2). 求
$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan(e^x) dx.$$

解 (1)利用定积分的换元法, 有

$$\begin{aligned}\int_{-a}^a f(x)g(x)\mathrm{d}x &= \int_{-a}^0 f(x)g(x)\mathrm{d}x + \int_0^a f(x)g(x)\mathrm{d}x \\&= - \int_a^0 f(-t)g(-t)\mathrm{d}t + \int_0^a f(x)g(x)\mathrm{d}x \\&\quad (\text{令 } x = -t) \\&= \int_0^a f(-x)g(x)\mathrm{d}x + \int_0^a f(x)g(x)\mathrm{d}x \\&= \int_0^a [f(-x) + f(x)]g(x)\mathrm{d}x \\&= A \int_0^a g(x)\mathrm{d}x\end{aligned}$$

(2) 设 $f(x) = \arctan(e^x)$, 则

$$[f(x) + f(-x)]' = \frac{e^x}{1 + e^{2x}} + \frac{-e^{-x}}{1 + e^{-2x}} = 0,$$

所以 $f(x) + f(-x)$ 恒为常数.

因为 $f(0) + f(-0) = \frac{\pi}{2}$, 所以

$$f(x) + f(-x) = \frac{\pi}{2}.$$

根据(1)的结论, 有

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\sin x| \arctan(e^x) dx = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} |\sin x| dx = \frac{\pi}{2}.$$

例8 若 $f(x)$ 是连续的周期函数, 周期为 T , 证明:

$$\int_a^{a+T} f(x) \mathrm{d}x = \int_0^T f(x) \mathrm{d}x,$$
$$\int_a^{a+nT} f(x) \mathrm{d}x = n \int_0^T f(x) \mathrm{d}x,$$

由此计算

$$\int_0^{n\pi} \sqrt{1 + \sin 2x} \mathrm{d}x.$$

证 方法一 令 $x = T + t$, 则

$$\int_T^{a+T} f(x) \mathrm{d}x = \int_0^a f(T+t) \mathrm{d}t = \int_0^a f(t) \mathrm{d}t,$$

所以

$$\begin{aligned} \int_a^{a+T} f(x) \mathrm{d}x &= \int_a^0 f(x) \mathrm{d}x + \int_0^T f(x) \mathrm{d}x + \int_T^{a+T} f(x) \mathrm{d}x \\ &= \int_a^0 f(x) \mathrm{d}x + \int_0^T f(x) \mathrm{d}x + \int_0^a f(x) \mathrm{d}x \\ &= \int_0^T f(x) \mathrm{d}x. \end{aligned}$$

方法二 令

$$\Phi(a) = \int_a^{a+T} f(x) \mathrm{d}x,$$

则有

$$\Phi'(a) = f(a+T) - f(a) = 0,$$

所以 $\Phi(a) = \Phi(0)$, 即

$$\int_a^{a+T} f(x) \mathrm{d}x = \int_0^T f(x) \mathrm{d}x.$$

由于 $\sqrt{1 + \sin 2x}$ 是以 π 为周期的函数, 则

$$\begin{aligned}\int_0^{n\pi} \sqrt{1 + \sin 2x} \mathrm{d}x &= n \int_0^{\pi} \sqrt{1 + \sin 2x} \mathrm{d}x \\ &= n \int_0^{\pi} |\sin x + \cos x| \mathrm{d}x \\ &= 2\sqrt{2}n.\end{aligned}$$

例9 设 $f(x) \in C(-\infty, +\infty)$, $F(x) = \int_0^x f(t-x)dt$,

求 $F'(x)$.

解 令 $t-x=u$.

当 $t=0$ 时, $u=-x$; 当 $t=x$ 时, $u=0$.

于是

$$F(x) = \int_0^x f(t-x)dt = \int_{-x}^0 f(u)du,$$

所以

$$F'(x) = 0 - f(-x) \cdot (-x)' = f(-x).$$

例10 已知 $f(x)$ 为连续函数, $F(x) = \int_0^x f(xt)dt$, 求 $F'(x)$.

解 当 $x \neq 0$ 时, 令 $u = xt$, 则 $F(x) = \frac{1}{x} \int_0^{x^2} f(u)du$

当 $x = 0$ 时, 有 $F(x) = \int_0^0 f(0t)dt = 0$.

所以

$$F(x) = \begin{cases} \frac{1}{x} \int_0^{x^2} f(u)du, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

$$\begin{aligned} \text{则 } F'(0) &= \lim_{x \rightarrow 0} \frac{F(x) - F(0)}{x} \\ &= \lim_{x \rightarrow 0} \frac{1}{x^2} \int_0^{x^2} f(u)du = \lim_{x \rightarrow 0} \frac{2xf(x^2)}{2x} = f(0). \end{aligned}$$

当 $x \neq 0$ 时,

$$\begin{aligned} F'(x) &= \left(\frac{1}{x} \int_0^{x^2} f(u) \mathrm{d}u \right)' \\ &= \frac{x \left[\int_0^{x^2} f(u) \mathrm{d}u \right]' - \int_0^{x^2} f(u) \mathrm{d}u}{x^2} \\ &= \frac{2x^2 f(x^2) - \int_0^{x^2} f(u) \mathrm{d}u}{x^2}. \end{aligned}$$

例11 设 $f(x)$ 可导, $F(x) = \int_0^x t^{n-1} f(x^n - t^n) dt$,

$f(0) = 0$, 求 $\lim_{x \rightarrow 0} \frac{F(x)}{x^{2n}}$.

$$\begin{aligned} \overset{\text{解}}{F(x)} &= \int_0^x t^{n-1} f(x^n - t^n) dt = \frac{1}{n} \int_0^x f(x^n - t^n) dt^n \\ &= -\frac{1}{n} \int_0^x f(x^n - t^n) d(x^n - t^n) \quad (\text{令 } u = x^n - t^n) \\ &= -\frac{1}{n} \int_{x^n}^0 f(u) du = \frac{1}{n} \int_0^{x^n} f(u) du, \end{aligned}$$

于是

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{F(x)}{x^{2n}} &= \lim_{x \rightarrow 0} \frac{\frac{1}{n} \int_0^{x^n} f(u) \mathrm{d}u}{x^{2n}} = \lim_{x \rightarrow 0} \frac{\frac{1}{n} f(x^n) \cdot nx^{n-1}}{2nx^{2n-1}} \\ &= \lim_{x \rightarrow 0} \frac{f(x^n)}{2nx^n} = \lim_{x \rightarrow 0} \frac{f(x^n) - f(0)}{2nx^n} \\ &= \frac{1}{2n} f'(0).\end{aligned}$$

例12 求连续函数 $f(x)$, 使它满足

$$\int_0^1 f(tx)dt = f(x) + x\sin x, \quad f(0) = 0.$$

解 设 $y = tx$, 有 $dt = \frac{1}{x}dy$.

当 $t = 0$ 时, $y = 0$; 当 $t = 1$ 时, $y = x$.

当 $x \neq 0$, 有

$$\frac{1}{x} \int_0^x f(y)dy = f(x) + x\sin x, \quad f(0) = 0.$$

所以

$$\int_0^x f(y)dy = xf(x) + x^2\sin x.$$

对上式两边求导, 有

$$f(x) = f(x) + xf'(x) + 2x\sin x + x^2\cos x \quad (x \neq 0).$$

整理可得

$$f'(x) = -2\sin x - x\cos x.$$

对上式两边在区间 $[0, x]$ 上积分, 有

$$\begin{aligned} f(x) - f(0) &= -2 \int_0^x \sin t dt - \int_0^x t \cos t dt \\ &= \cos x - x \sin x - 1, \end{aligned}$$

则所求函数为

$$f(x) = \cos x - x \sin x - 1.$$

例13 设 $f(x) \in C[0, 1]$, 且 $f(x)$ 在 $[0, 1]$ 上单调减少, 证明: $\forall q \in (0, 1)$, 成立不等式

$$\int_0^q f(x) \mathrm{d}x \geq q \int_0^1 f(x) \mathrm{d}x.$$

证 令 $x = qt$.

当 $x = 0$ 时, $t = 0$; 当 $x = q$ 时, $t = 1$.

于是

$$\int_0^q f(x) \mathrm{d}x = \int_0^1 f(qt) \cdot q \mathrm{d}t = q \int_0^1 f(qt) \mathrm{d}t = q \int_0^1 f(qx) \mathrm{d}x,$$

于是

$$\begin{aligned}\int_0^q f(x) \mathrm{d}x - q \int_0^1 f(x) \mathrm{d}x &= q \int_0^1 f(qx) \mathrm{d}x - q \int_0^1 f(x) \mathrm{d}x \\ &= q \int_0^1 [f(qx) - f(x)] \mathrm{d}x.\end{aligned}$$

因为 $f(x)$ 在 $[0, 1]$ 上单调减少, 又 $0 \leq qx \leq x \leq 1$,
故 $f(qx) \geq f(x)$, 从而

$$\int_0^1 [f(qx) - f(x)] \mathrm{d}x \geq 0.$$

即

$$\int_0^q f(x) \mathrm{d}x \geq q \int_0^1 f(x) \mathrm{d}x.$$

3.2 分部积分法

由导数的乘法公式, 反过来即得分部积分公式.

设 $u(x), v(x)$ 可导, 若 $\int u'(x)v(x)\mathrm{d}x$ 存在, 则

$$\int u(x)v'(x)\mathrm{d}x = u(x)v(x) - \int v(x)u'(x)\mathrm{d}x.$$

该公式也常写成如下形式:

$$\int u\mathrm{d}v = uv - \int v\mathrm{d}u.$$

例1 求 $\int x e^{-x} dx$.

解 令 $u = x, v' = e^{-x}$, 则

$$\begin{aligned}\int x e^{-x} dx &= - \int x de^{-x} \\ &= -x e^{-x} + \int e^{-x} dx \\ &= -x e^{-x} - e^{-x} + C.\end{aligned}$$

例2 求 $\int x^2 \sin x dx$.

解 令 $u = x^2, v' = \sin x$, 则

$$\begin{aligned}\int x^2 \sin x dx &= - \int x^2 d\cos x \\&= -x^2 \cos x + 2 \int x \cos x dx \\&= -x^2 \cos x + 2 \int x d\sin x \\&= -x^2 \cos x + 2x \sin x - 2 \int \sin x dx \\&= -x^2 \cos x + 2x \sin x + 2 \cos x + C.\end{aligned}$$

⊙选择 u 和 dv 的原则:

(1) v 要容易求出;

(2) $\int vdu$ 要比 $\int u dv$ 容易求出.

例3 求 $\int \ln x dx$.

解 令 $u = \ln x, v' = 1$, 则

$$\begin{aligned}\int \ln x dx &= x \ln x - \int x d \ln x \\ &= x \ln x - \int x \cdot \frac{1}{x} dx \\ &= x \ln x - x + C.\end{aligned}$$

例4 求 $\int \arctan x dx$.

解 令 $u = \arctan x$, $v' = 1$, 则

$$\begin{aligned}\int \arctan x dx &= x \cdot \arctan x - \int x d(\arctan x) \\&= x \cdot \arctan x - \int \frac{x}{1+x^2} dx \\&= x \cdot \arctan x - \frac{1}{2} \int \frac{1}{1+x^2} d(1+x^2) \\&= x \cdot \arctan x - \frac{1}{2} \ln|1+x^2| + C.\end{aligned}$$

例5 求 $\int \frac{x}{\cos^2 x} dx$.

解 令 $u = x, v' = \frac{1}{\cos^2 x}$, 则

$$\begin{aligned}\int \frac{x}{\cos^2 x} dx &= \int x \tan x \\ &= x \cdot \tan x - \int \tan x dx \\ &= x \cdot \tan x + \ln |\cos x| + C.\end{aligned}$$

例6 求 $\int e^{ax} \cos bx dx$.

解 根据分部积分法

$$\begin{aligned}\int e^{ax} \cos bx dx &= \frac{1}{a} \int \cos bx de^{ax} \\ &= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} \int e^{ax} \sin bx dx,\end{aligned}$$

再次利用分部积分法

$$\begin{aligned}\text{上式} &= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} \int \sin bx de^{ax} \\ &= \frac{1}{a} e^{ax} \cos bx + \frac{b}{a^2} e^{ax} \sin bx - \frac{b^2}{a^2} \int e^{ax} \cos bx dx,\end{aligned}$$

可得

$$\int e^{ax} \cos bx dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} + C.$$

例7 求 $\int \sin(\ln x) dx$.

解

$$\begin{aligned}\int \sin(\ln x) dx &= x \sin(\ln x) - \int \cos(\ln x) dx \\ &= x \sin(\ln x) - x \cos(\ln x) - \int \sin(\ln x) dx\end{aligned}$$

所以 $\int \sin(\ln x) dx = \frac{x}{2} [\sin(\ln x) - \cos(\ln x)] + C.$

例8 求 $\int xe^x \cos x dx$, $\int xe^x \sin x dx$.

解 利用分部积分

$$\int xe^x \cos x dx = \int x \cos x de^x = x \cos x e^x - \int (x \cos x)' \cdot e^x dx$$

$$= x \cos x e^x - \int [\cos x - x \sin x] \cdot e^x dx$$

$$= x \cos x e^x - \underbrace{\int \cos x e^x dx} + \underbrace{\int x \sin x e^x dx}$$

$$\int xe^x \sin x dx = \int x \sin x de^x = x \sin x e^x - \int (x \sin x)' \cdot e^x dx$$

$$= x \sin x e^x - \int [\sin x + x \cos x] \cdot e^x dx$$

$$= x \sin x e^x - \underbrace{\int \sin x e^x dx} - \underbrace{\int x \cos x e^x dx},$$

根据上面两个等式可得 $\int xe^x \cos x dx$, $\int xe^x \sin x dx$ 的值.

例9 求 $I_n = \int \frac{dx}{(x^2 + a^2)^n}$ (n 是自然数).

解 根据分部积分, 设 $u = \frac{1}{(x^2 + a^2)^n}$, $dv = dx$, 则

$$du = \frac{-2nx}{(x^2 + a^2)^{n+1}} dx, v = x, \text{ 所以}$$

$$\begin{aligned} I_n &= \frac{x}{(x^2 + a^2)^n} + \int \frac{2nx^2}{(x^2 + a^2)^{n+1}} dx \\ &= \frac{x}{(x^2 + a^2)^n} + 2n \int \frac{x^2 + a^2 - a^2}{(x^2 + a^2)^{n+1}} dx \\ &= \frac{x}{(x^2 + a^2)^n} + 2n \int \frac{dx}{(x^2 + a^2)^n} - 2na^2 \int \frac{dx}{(x^2 + a^2)^{n+1}} \end{aligned}$$

即得
$$I_n = \frac{x}{(x^2 + a^2)^n} + 2nI_n - 2na^2 I_{n+1}.$$

所以

$$I_{n+1} = \frac{1}{2na^2} \cdot \frac{x}{(x^2 + a^2)^n} + \frac{2n-1}{2na^2} I_n.$$

并且

$$I_1 = \frac{1}{a} \arctan \frac{x}{a} + C.$$

以此作递推公式, 计算出 I_n .

例10 求 $\int x^n e^{-x} dx$ (n 是自然数).

解 当 $n = 1$ 时,

$$I_1 = \int x e^{-x} dx = -(x + 1)e^{-x} + C.$$

当 $n > 1$ 时,

$$\begin{aligned} I_n &= \int x^n e^{-x} dx = \int x^n d(-e^{-x}) \\ &= -x^n e^{-x} + \int e^{-x} d(x^n) \\ &= -x^n e^{-x} + n \int x^{n-1} e^{-x} dx \\ &= -x^n e^{-x} + n I_{n-1}. \end{aligned}$$

三个重要不定积分

例1 求 $\int \sqrt{a^2 - x^2} dx$.

$$\begin{aligned}\text{解 } \int \sqrt{a^2 - x^2} dx &= x\sqrt{a^2 - x^2} - \int x d\sqrt{a^2 - x^2} \\&= x\sqrt{a^2 - x^2} - \int x \cdot \frac{-x}{\sqrt{a^2 - x^2}} dx \\&= x\sqrt{a^2 - x^2} - \int \frac{a^2 - x^2 - a^2}{\sqrt{a^2 - x^2}} dx \\&= x\sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2} dx + \int \frac{a^2}{\sqrt{a^2 - x^2}} dx \\&= x\sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2} dx + a^2 \arcsin \frac{x}{a},\end{aligned}$$

$$\text{则 } \int \sqrt{a^2 - x^2} dx = \frac{1}{2} \left(x\sqrt{a^2 - x^2} + a^2 \arcsin \frac{x}{a} \right) + C.$$

例2 求 $\int \sqrt{x^2 - a^2} dx$.

解 $\int \sqrt{x^2 - a^2} dx = x\sqrt{x^2 - a^2} - \int x d\sqrt{x^2 - a^2}$

$$= x\sqrt{x^2 - a^2} - \int x \cdot \frac{x}{\sqrt{x^2 - a^2}} dx$$

$$= x\sqrt{x^2 - a^2} - \int \frac{x^2 - a^2 + a^2}{\sqrt{x^2 - a^2}} dx$$

$$= x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2} dx - \int \frac{a^2}{\sqrt{x^2 - a^2}} dx$$

$$= x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2} dx - a^2 \ln |x + \sqrt{x^2 - a^2}|,$$

$$\text{则 } \int \sqrt{x^2 - a^2} dx = \frac{1}{2} \left(x\sqrt{x^2 - a^2} - a^2 \ln |x + \sqrt{x^2 - a^2}| \right) + C.$$

例3 求 $\int \sqrt{x^2 + a^2} dx$.

解 $\int \sqrt{x^2 + a^2} dx = x\sqrt{x^2 + a^2} - \int x d\sqrt{x^2 + a^2}$

$$= x\sqrt{x^2 + a^2} - \int x \cdot \frac{x}{\sqrt{x^2 + a^2}} dx$$

$$= x\sqrt{x^2 + a^2} - \int \frac{x^2 + a^2 - a^2}{\sqrt{x^2 + a^2}} dx$$

$$= x\sqrt{x^2 + a^2} - \int \sqrt{x^2 + a^2} dx + \int \frac{a^2}{\sqrt{x^2 + a^2}} dx$$

$$= x\sqrt{x^2 + a^2} - \int \sqrt{x^2 + a^2} dx + a^2 \ln |x + \sqrt{x^2 + a^2}|,$$

则 $\int \sqrt{x^2 + a^2} dx = \frac{1}{2} \left(x\sqrt{a^2 + x^2} + a^2 \ln |x + \sqrt{x^2 + a^2}| \right) + C.$

定积分的分部积分法

设函数 $u = u(x)$, $v = v(x)$ 在区间 $[a, b]$ 上有可积的导函数 $u'(x)$, $v'(x)$, 则

$$\int_a^b uv' dx = [uv] \Big|_a^b - \int_a^b u'v dx.$$

上式可写成

$$\int_a^b u dv = [uv] \Big|_a^b - \int_a^b v du.$$

例1 求 $\int_0^1 e^{\sqrt{x}} dx$.

解 先用换元法, 令 $x = t^2$, $dx = 2t dt$;
并且当 $x = 0$ 时, $t = 0$; 当 $x = 1$ 时, $t = 1$.
于是

$$\int_0^1 e^{\sqrt{x}} dx = 2 \int_0^1 t e^t dt.$$

再用分部积分法. 设 $u = t$, $dv = e^t dt$, 则

$$\int_0^1 t e^t dt = [t e^t] \Big|_0^1 - \int_0^1 e^t dt = e - [e^t] \Big|_0^1 = 1.$$

例2 求 $\int_0^3 \arcsin \sqrt{\frac{x}{1+x}} dx$.

解 令 $\sin^2 t = \frac{x}{1+x}$, 所以 $x = \tan^2 t$, $dx = d(\tan^2 t)$.

当 $x = 0$ 时, $t = 0$; $x = 3$ 时, $t = \frac{\pi}{3}$.

于是, 根据分部积分

$$\begin{aligned} I &= \int_0^{\frac{\pi}{3}} t d(\tan^2 t) = [t \cdot \tan^2 t] \Big|_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \tan^2 t dt \\ &= \frac{\pi}{3} \cdot 3 - \int_0^{\frac{\pi}{3}} (\sec^2 t - 1) dt \\ &= \pi - [\tan t - t] \Big|_0^{\frac{\pi}{3}} = \pi - \sqrt{3} + \frac{\pi}{3} = \frac{4}{3}\pi - \sqrt{3}. \end{aligned}$$

例3 求 $\int_{e^{-1}}^e |\ln x| dx$.

解 根据分部积分, 有

$$\begin{aligned} \text{原式} &= \int_{e^{-1}}^1 (-\ln x) dx + \int_1^e \ln x dx \\ &= \left([-x \ln x] \Big|_{e^{-1}}^1 + \int_{e^{-1}}^1 dx \right) + \left([x \ln x] \Big|_1^e - \int_1^e dx \right) \\ &= 2 - 2e^{-1}. \end{aligned}$$

例4 求 $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$, 其中, n 为自然数.

解 利用分部积分. 设 $u = \sin^{n-1} x$, $dv = \sin x dx$, 得

$$\begin{aligned} I_n &= - \int_0^{\frac{\pi}{2}} \sin^{n-1} x d\cos x \\ &= [-\sin^{n-1} x \cos x] \Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x d\sin^{n-1} x \\ &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx \end{aligned}$$

于是

$$I_n = \frac{n-1}{n} I_{n-2},$$

容易算出

$$I_0 = \int_0^{\frac{\pi}{2}} (\sin x)^0 dx = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2},$$

$$I_1 = \int_0^{\frac{\pi}{2}} \sin x dx = [-\cos x]_0^{\frac{\pi}{2}} = 1.$$

由此得到

$$I_{2n} = \frac{(2n-1)!!}{(2n)!!} \frac{\pi}{2},$$

$$I_{2n+1} = \frac{(2n)!!}{(2n+1)!!}.$$

例5 求 $I_m = \int_0^{\pi} x \sin^m x dx$ (m 是自然数).

解 利用换元法, 令 $x = \pi - t$. 则

$$\begin{aligned} I_m &= \int_0^{\pi} x \sin^m x dx = \int_{\pi}^0 (\pi - t) \sin^m(\pi - t) (-dt) \\ &= \int_0^{\pi} (\pi - t) \sin^m t dt = \pi \int_0^{\pi} \sin^m t dt - \int_0^{\pi} t \sin^m t dt \\ &= 2\pi \int_0^{\frac{\pi}{2}} \sin^m t dt - I_m \quad (**) \end{aligned}$$

所以

$$I_m = \pi \int_0^{\frac{\pi}{2}} \sin^m x dx = \begin{cases} \frac{(m-1)!!}{m!!} \cdot \pi, & m = 1, 3, 5, \dots \\ \frac{(m-1)!!}{m!!} \cdot \frac{\pi^2}{2}, & m = 2, 4, 6, \dots \end{cases}$$

关于(**)式, 首先

$$\int_0^{\pi} \sin^m t dt = \int_0^{\frac{\pi}{2}} \sin^m t dt + \int_{\frac{\pi}{2}}^{\pi} \sin^m t dt,$$

其中第二项, 令 $t = \pi - x$, 则当 $t = \frac{\pi}{2}$ 时, 有 $x = \frac{\pi}{2}$; 当 $t = \pi$ 时, 有 $x = 0$. 所以

$$\begin{aligned} \int_{\frac{\pi}{2}}^{\pi} \sin^m t dt &= \int_{\frac{\pi}{2}}^0 \sin^m(\pi - x) d(\pi - x) \\ &= \int_{\frac{\pi}{2}}^0 \sin^m x (-dx) = \int_0^{\frac{\pi}{2}} \sin^m x dx = \int_0^{\frac{\pi}{2}} \sin^m t dt. \end{aligned}$$

例6 求 $I_{m,n} = \int_0^1 x^m(1-x)^n dx$, (m, n 是自然数).

证 利用分部积分公式,

$$\begin{aligned} I_{m,n} &= \frac{1}{m+1} \int_0^1 (1-x)^n dx^{m+1} \\ &= \frac{1}{m+1} [(1-x)^n x^{m+1}] \Big|_0^1 + \frac{n}{m+1} \int_0^1 x^{m+1} (1-x)^{n-1} dx \\ &= \frac{n}{m+1} \int_0^1 x^{m+1} (1-x)^{n-1} dx \\ &= \frac{n}{m+1} I_{m+1, n-1}, \end{aligned}$$

于是

$$\begin{aligned} I_{m,n} &= \frac{n}{m+1} I_{m+1,n-1} \\ &= \frac{n}{m+1} \cdot \frac{n-1}{m+2} \cdot I_{m+2,n-2} \\ &= \frac{n}{m+1} \cdot \frac{n-1}{m+2} \cdots \frac{n-(n-1)}{m+n} I_{m+n,0} \\ &= \frac{n!}{(m+1)(m+2) \cdots (m+n)} \int_0^1 x^{m+n} \mathrm{d}x \\ &= \frac{n!}{(m+1)(m+2) \cdots (m+n)} \frac{1}{m+n+1}. \end{aligned}$$

例7 设 $f''(x) \in C[a, b]$, 证明:

$$\int_a^b f(x) \mathrm{d}x = \frac{b-a}{2} [f(a) + f(b)] \\ + \frac{1}{2} \int_a^b f''(x)(x-a)(x-b) \mathrm{d}x.$$

证 利用分部积分法, 则

$$\begin{aligned} & \int_a^b f''(x)(x-a)(x-b) \mathrm{d}x \\ &= (x-a)(x-b)f'(x) \Big|_a^b - \int_a^b f'(x)(2x-a-b) \mathrm{d}x \\ &= (a+b-2x)f(x) \Big|_a^b + 2 \int_a^b f(x) \mathrm{d}x \\ &= (a-b)f(b) - (b-a)f(a) + 2 \int_a^b f(x) \mathrm{d}x. \end{aligned}$$

所以

$$\begin{aligned}\int_a^b f(x) \mathrm{d}x &= \frac{b-a}{2} [f(a) + f(b)] \\ &\quad + \frac{1}{2} \int_a^b f''(x)(x-a)(x-b) \mathrm{d}x.\end{aligned}$$

例8. 设 $f(x)$ 单调增加且有连续的导数, 又 $f(0) = 0$, $f(a) = b$, $g(x)$ 是 $f(x)$ 的反函数, 试证

$$\int_0^a f(x) \mathrm{d}x + \int_0^b g(x) \mathrm{d}x = ab.$$

证 利用变量代换 $x = g(y)$, 则

$$\begin{aligned} \int_0^b g(y) \mathrm{d}y &= \int_0^a x \mathrm{d}f(x) \\ &= [xf(x)] \Big|_{x=0}^a - \int_0^a f(x) \mathrm{d}x \\ &= af(a) - \int_0^a f(x) \mathrm{d}x. \end{aligned}$$

例9 设 $f(x)$ 为连续函数, 证明:

$$\int_0^x f(t)(x-t)dt = \int_0^x \left(\int_0^t f(u)du \right) dt.$$

证 令 $F(t) = \int_0^t f(u)du$, 则

$$\begin{aligned} \int_0^x \left(\int_0^t f(u)du \right) dt &= \int_0^x F(t)dt = [tF(t)] \Big|_0^x - \int_0^x tF'(t)dt \\ &= xF(x) - \int_0^x tf(t)dt \\ &= x \int_0^x f(u)du - \int_0^x tf(t)dt \\ &= \int_0^x xf(t)dt - \int_0^x tf(t)dt = \int_0^x (x-t)f(t)dt. \end{aligned}$$

例10 已知 $f(x)$ 在 $[0, 1]$ 上连续, 且

$$f(x) = \sqrt{x} - 2x \int_0^1 f(x) \mathrm{d}x$$

求 $f(x)$.

解 注意 $\int_0^1 f(x) \mathrm{d}x$ 是一个常数, 设 $\int_0^1 f(x) \mathrm{d}x = A$, 则

$$\begin{aligned} A &= \int_0^1 f(x) \mathrm{d}x = \int_0^1 [\sqrt{x} - 2xA] \mathrm{d}x \\ &= \left[\frac{2}{3} x^{\frac{3}{2}} \right] \bigg|_0^1 - A[x^2] \bigg|_0^1 = \frac{2}{3} - A \end{aligned}$$

所以 $A = \frac{1}{3}$, $f(x) = \sqrt{x} - \frac{2}{3}x$.

问题1: 用微分学方法, 证明与定积分有关的等式或不等式.

例1 证明: 当 $a \in \left[0, \frac{\pi}{2}\right]$ 时, 有

$$\int_0^{\sin^2 a} \arcsin \sqrt{t} dt + \int_0^{\cos^2 a} \arccos \sqrt{t} dt = \frac{\pi}{4}.$$

证 令 $f(x) = \int_0^{\sin^2 x} \arcsin \sqrt{t} dt + \int_0^{\cos^2 x} \arccos \sqrt{t} dt.$

则

$$\begin{aligned}f'(x) &= \arcsin(\sin x) \cdot 2 \sin x \cos x \\&\quad + \arccos(\cos x) \cdot (-2 \cos x \sin x) \\&= x \sin 2x - x \sin 2x = 0,\end{aligned}$$

故 $f(x) \equiv C$. 又

$$\begin{aligned}C &= f\left(\frac{\pi}{4}\right) = \int_0^{\frac{1}{2}} \arcsin \sqrt{t} dt + \int_0^{\frac{1}{2}} \arccos \sqrt{t} dt \\&= \int_0^{\frac{1}{2}} (\arcsin \sqrt{t} + \arccos \sqrt{t}) dt = \int_0^{\frac{1}{2}} \frac{\pi}{2} dt = \frac{\pi}{4}.\end{aligned}$$

例2 设 $f(x)$ 在 $[a, b]$ 上是单调增加的, 且二阶可导, $f''(x) > 0$, 求证:

$$\int_a^b f(x) \mathrm{d}x < (b-a) \frac{f(a) + f(b)}{2}.$$

证 考虑函数

$$F(x) = (x - a) \frac{f(a) + f(x)}{2} - \int_a^x f(t) dt,$$

因为

$$\begin{aligned} F'(x) &= \frac{f(a) + f(x)}{2} + (x - a) \frac{f'(x)}{2} - f(x) \\ &= \frac{f(a) - f(x)}{2} + (x - a) \frac{f'(x)}{2} \\ &\quad (\text{因为 } f(a) - f(x) = f'(\xi)(a - x), \xi \in (a, x)) \\ &= \frac{1}{2}(a - x)f'(\xi) + \frac{1}{2}(x - a)f'(x) \\ &= \frac{1}{2}(a - x)[f'(x) - f'(\xi)] > 0, \end{aligned}$$

其中, 因为 $f''(x) > 0$, 所以 $f'(x)$ 单调增加.

所以, $F(x)$ 单调增加. 于是 $F(b) > F(a) = 0$.

例3 设 $f(x) \in C[a, b]$, 且单调增加, 求证:

$$\int_a^b x f(x) dx > \frac{a+b}{2} \int_a^b f(x) dx.$$

证 考虑函数

$$F(x) = \frac{a+x}{2} \int_a^x f(t) dt - \int_a^x t f(t) dt,$$

因为

$$\begin{aligned} F'(x) &= \frac{1}{2} \int_a^x f(t) dt + \frac{a+x}{2} f(x) - x f(x) \\ &= \frac{1}{2} \int_a^x f(t) dt + \frac{1}{2} (a-x) f(x) \\ &= \frac{1}{2} f(\xi) (x-a) - \frac{1}{2} (a-x) f(x) \\ &= \frac{1}{2} [f(\xi) - f(x)] (x-a) < 0, \quad \xi \in (a, x), \end{aligned}$$

所以, $F(x)$ 单调减. 所以 $F(b) < F(a) = 0$.

例4 设 $f(x) \in C[0, 1]$, 且 $f(x)$ 在 $[0, 1]$ 上单调减少, 证明:
 $\forall q \in (0, 1)$, 成立不等式

$$\int_0^q f(x) dx \geq q \int_0^1 f(x) dx.$$

证 要证不等式即

$$q \int_0^1 f(t) dt - 1 \cdot \int_0^q f(t) dt \leq 0,$$

考虑函数

$$F(x) = q \int_0^x f(t) dt - x \int_0^q f(t) dt, \quad x > q,$$

即要证 $F(1) \leq 0$. 因为 $1 > q$, 所以考虑 $F(x)$ 的定义域为 $x > q$.

因为

$$\begin{aligned}F'(x) &= qf(x) - \int_0^q f(t)\mathrm{d}t \\&= f(x) \int_0^q \mathrm{d}t - \int_0^q f(t)\mathrm{d}t \\&= \int_0^q f(x)\mathrm{d}t - \int_0^q f(t)\mathrm{d}t \\&= \int_0^q [f(x) - f(t)]\mathrm{d}t \leqslant 0\end{aligned}$$

其中, 因为 $t \in [0, q]$, $x > q$, 所以 $x > t$. 则 $f(x) \leqslant f(t)$.
所以

$$F(1) \leqslant F(q) = 0,$$

得证.

例5. 设 $f(x)$ 在 $[a, b]$ 上连续, $f(x) > 0$, 试证:

$$\int_a^b f(x) \mathrm{d}x \cdot \int_a^b \frac{1}{f(x)} \mathrm{d}x \geqslant (b-a)^2, \quad (b > a)$$

证 考虑函数

$$F(x) = \int_a^x f(t)dt \cdot \int_a^x \frac{1}{f(t)}dt - (x-a)^2.$$

则有

$$\begin{aligned} F'(x) &= f(x) \cdot \int_a^x \frac{1}{f(t)}dt + \frac{1}{f(x)} \cdot \int_a^x f(t)dt - 2(x-a) \\ &= \int_a^x \frac{f(x)}{f(t)}dt + \int_a^x \frac{f(t)}{f(x)}dt - \int_a^x 2dt \\ &= \int_a^x \left[\frac{f(x)}{f(t)} + \frac{f(t)}{f(x)} - 2 \right] dt \\ &\geq 0 \end{aligned}$$

所以 $F(x)$ 单调增加, $F(b) \geq F(a) = 0$.

例6. 设 $f(x)$ 在 $[0, 1]$ 上可导, 且当 $x \in (0, 1)$ 时,
 $0 < f'(x) < 1$, $f(0) = 0$, 试证:

$$\left(\int_0^1 f(x) dx \right)^2 > \int_0^1 f^3(x) dx.$$

证 考虑函数

$$F(x) = \left(\int_0^x f(t) dt \right)^2 - \int_0^x f^3(t) dt.$$

则有 $F(0) = 0$.

因为

$$\begin{aligned} F'(x) &= 2f(x) \int_0^x f(t) \mathrm{d}t - f^3(x) \\ &= f(x) \left[2 \int_0^x f(t) \mathrm{d}t - f^2(x) \right] \end{aligned}$$

根据 $f(0) = 0$, $0 < f'(x) < 1$. 所以, 当 $x \in (0, 1)$ 时, 有 $f(x) > 0$.

设 $g(x) = 2 \int_0^x f(t) \mathrm{d}t - f^2(x)$, 则有 $g(0) = 0$, 且

$$g'(x) = 2f(x) - 2f(x)f'(x) = 2f(x)[1 - f'(x)] > 0,$$

所以, $g(x) > g(0) = 0$.

于是, 当 $x \in (0, 1)$ 时, $F'(x) > 0$. 所以

$$F(1) > F(0) = 0.$$

例7. 设 $f(x)$ 为 $(-\infty, +\infty)$ 上连续的奇函数, 且以 $2T$ 为周期的周期函数, 证明 $F(x) = \int_0^x f(t)dt$ 也为以 $2T$ 为周期的周期函数.

解 方法一 因为

$$F(x + 2T) = \int_0^{x+2T} f(t)dt = \int_0^x f(t)dt + \int_x^{x+2T} f(t)dt,$$

其中, 因为 $f(x)$ 为奇函数, 且以 $2T$ 为周期, 所以

$$\int_x^{x+2T} f(t)dt = \int_{-T}^T f(t)dt = 0.$$

所以

$$F(x + 2T) = \int_0^x f(t)dt = F(x).$$

方法二 要证 $\int_0^{x+2T} f(t)dt = \int_0^x f(t)dt$, 令

$$G(x) = \int_0^{x+2T} f(t)dt - \int_0^x f(t)dt,$$

因为

$$G'(x) = f(x+2T) - f(x) = 0,$$

所以

$$\begin{aligned} G(x) &= G(0) = \int_0^{0+2T} f(t)dt - \int_0^0 f(t)dt \\ &= \int_0^{0+2T} f(t)dt = \int_{-T}^T f(t)dt = 0. \end{aligned}$$

问题2: 利用微分中值定理, 泰勒公式等, 证明与定积分有关的等式和不等式.

(1). 当问题条件涉及到二阶或更高阶导数时, 可利用泰勒公式处理.

(2). 在使用泰勒公式时, 一般是将函数在一些比较特殊的点. 例如, 区间中点, 边界点, 函数零点等处做展开.

例1 设 $f(x)$ 在 $[a, b]$ 上可导, 且 $f'(x) \leq M$, $f(a) = 0$, 试证:

$$\int_a^b f(x) \mathrm{d}x \leq \frac{1}{2}M(b-a)^2.$$

证 根据拉格朗日中值定理, $\forall x \in [a, b]$, $\exists \xi \in (a, b)$, 使得

$$f(x) = f(x) - f(a) = f'(\xi)(x - a).$$

则有

$$\begin{aligned} \int_a^b f(x) \mathrm{d}x &= \int_a^b f'(\xi)(x - a) \mathrm{d}x \leq M \int_a^b (x - a) \mathrm{d}x \\ &= \frac{1}{2}M(b - a)^2. \end{aligned}$$

例2 设 $f(x) \geq 0$, $f''(x) < 0$, 证明: $\forall x \in [a, b]$,

$$f(x) \leq \frac{2}{b-a} \int_a^b f(t) dt.$$

证 $\forall x, t \in [a, b]$, 有

$$f(x) = f(t) + f'(t)(x-t) + \frac{f''(\xi)}{2!}(x-t)^2, \quad \xi \text{ 在 } x \text{ 与 } t \text{ 之间}.$$

因为 $\frac{f''(\xi)}{2!}(x-t)^2 \leq 0$, 所以

$$f(x) \leq f(t) + f'(t)(x-t), \quad \forall x \in [a, b].$$

所以

$$\begin{aligned} f(x) \cdot (b-a) &= \int_a^b f(x) dt \\ &\leq \int_a^b f(t) dt + \int_a^b f'(t)(x-t) dt. \end{aligned}$$

其中,

$$\begin{aligned}\int_a^b f'(t)(x-t)\mathrm{d}t &= [f(t)(x-t)]\Big|_{t=a}^{t=b} + \int_a^b f(t)\mathrm{d}t \\ &= f(b)(x-b) - f(a)(x-a) + \int_a^b f(t)\mathrm{d}t \\ &\leq \int_a^b f(t)\mathrm{d}t.\end{aligned}$$

$$\text{所以 } f(x) \cdot (b-a) \leq 2 \int_a^b f(t)\mathrm{d}t.$$

例3 设 $f(x)$ 在 $[a, b]$ 上二阶连续可微, $f\left(\frac{a+b}{2}\right) = 0$,

试证:

$$\left| \int_a^b f(x) \mathrm{d}x \right| \leq M \frac{(b-a)^3}{24}.$$

其中, $M = \max_{a \leq x \leq b} |f''(x)|$.

证 将 $f(x)$ 在 $x = \frac{a+b}{2}$ 处用泰勒公式展开, 则

$$\begin{aligned} f(x) &= f\left(\frac{a+b}{2}\right) + f'\left(\frac{a+b}{2}\right) \left(x - \frac{a+b}{2}\right) \\ &\quad + \frac{1}{2!} f''(\xi) \left(x - \frac{a+b}{2}\right)^2. \end{aligned}$$

则

$$\begin{aligned}\int_a^b f(x) \mathrm{d}x &= \int_a^b f' \left(\frac{a+b}{2} \right) \left(x - \frac{a+b}{2} \right) \mathrm{d}x \\ &\quad + \frac{1}{2!} \int_a^b f''(\xi) \left(x - \frac{a+b}{2} \right)^2 \mathrm{d}x \\ &= f' \left(\frac{a+b}{2} \right) \int_a^b \left(x - \frac{a+b}{2} \right) \mathrm{d}x \\ &\quad + \int_a^b \frac{f''(\xi)}{2!} \left(x - \frac{a+b}{2} \right)^2 \mathrm{d}x \\ &= 0 + \int_a^b \frac{f''(\xi)}{2!} \left(x - \frac{a+b}{2} \right)^2 \mathrm{d}x,\end{aligned}$$

所以

$$\begin{aligned} \left| \int_a^b f(x) \mathrm{d}x \right| &= \left| \int_a^b \frac{f''(\xi)}{2!} \left(x - \frac{a+b}{2} \right)^2 \mathrm{d}x \right| \\ &\leq \int_a^b \left| \frac{f''(\xi)}{2!} \right| \left(x - \frac{a+b}{2} \right)^2 \mathrm{d}x \\ &\leq \frac{M}{2!} \int_a^b \left(x - \frac{a+b}{2} \right)^2 \mathrm{d}x \\ &= M \frac{(b-a)^3}{24}. \end{aligned}$$

例4 设 $f(x)$ 二阶可导, 且 $f''(x) \geq 0$, $\varphi(t) \in C[a, b]$ 上可导, 证明:

$$\frac{1}{b-a} \int_a^b f[\varphi(t)] \mathrm{d}t \geq f \left[\frac{1}{b-a} \int_a^b \varphi(t) \mathrm{d}t \right].$$

证 根据泰勒公式, 有

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(\xi)}{2}(x - x_0)^2,$$

其中, ξ 介于 x_0 与 x 之间. 因为 $f''(x) \geq 0$, 所以

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0).$$

取 $x_0 = \frac{1}{b-a} \int_a^b \varphi(t) dt$, $x = \varphi(t)$. 在上式从 a 到 b 积分, 并除以 $b-a$, 得

$$\frac{1}{b-a} \int_a^b f(\varphi(t)) dt \geq \frac{1}{b-a} \int_a^b f(x_0) dt + \frac{f'(x_0)}{b-a} \int_a^b (\varphi(t) - x_0) dt.$$

得证.